

BLACK HAT EUROPE ARSENAL — SUBMISSION PROPOSAL

Tinker Tailor LLM Spy: Reconstructing "Deleted" Chats & Hijacking Sessions from Chromium LevelDB Caches

Track: Forensic / Incident Response / Malware Analysis	Format: 30-40 Minute Fast Track / Demo
Target OS: Windows, macOS, Linux	Targets: ChatGPT, Claude, Gemini, DeepSeek, Perplexity, Cursor, Copilot
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I. Executive Abstract (Proposal Submission)

AI coding assistants and browser LLM portals have created a critical security blind spot: employees pasting proprietary source code, credentials, and PII into LLMs, as well as adversaries leveraging AI tools during post-exploitation. **Verity** is an open-source, sub-50ms forensic carving and threat hunting engine designed to extract, reconstruct, and inspect LLM telemetry from local volatile artifacts. Unlike traditional forensic tools (such as Autopsy or Magnet AXIOM) that fail on non-standard database structures, Verity natively parses Chrome/Edge V8 IndexedDB LevelDB SSTables, Claude TipTap draft keystroke buffers, and IDE JSONL transcripts.

II. Key Technical Innovations & Threat Vector

- **Sub-50ms Execution (41ms Infostealer PoC):** Demonstrates how a headless Python script (verity_stealer.py) harvests 485+ carved prompts and session cookies before security tools register file modifications.
- **V8 & TipTap Keystroke Reconstruction:** Reconstructs deleted ChatGPT multi-turn conversation trees and unsubmitted Claude keystroke drafts.
- **7-Platform Unified Support:** Native carvers for ChatGPT, Claude AI, Gemini, DeepSeek, Perplexity, Cursor IDE, and VS Code Copilot.
- **Real-time Shadow AI DLP Engine:** Live regex inspection alerting on leaked AWS keys, Slack tokens, OpenAI API keys, and PII directly inside carved chat bubbles.

III. Technical Advantage Matrix vs. Legacy Forensic Suites

Capability / Feature	Autopsy / Magnet AXIOM	Verity Forensics
ChatGPT V8 Tree Recovery	Generic Strings Only (Unstructured)	100% Tree & Node Recovery
Claude TipTap Keystroke Drafts	0 Records Recognized	Prefix Deduplicated Recovery
Cursor & Copilot JSONL Parsing	Unsupported Schema	Native Agent Crawler
Secret Leak Detection & DLP	Basic YARA Regex	Real-time Scanner + UI Badges
Execution Speed	Minutes to Hours	51.9 milliseconds

IV. Compact 30-40 Minute Demonstration Schedule

Time	Topic / Demo Step	Key Arsenal Takeaway
00:00 - 05:00	Introduction & Threat Landscape	Browser & IDE LLM data persistence blind spots.
05:00 - 15:00	Sub-50ms Infostealer Vector Demo	Executing 41ms stealer harvesting 485+ prompts.
15:00 - 28:00	LevelDB V8 & TipTap Carving Deep Dive	Varint decoding & unsubmitted keystroke recovery.
28:00 - 35:00	Verity Dashboard & DLP Live Demo	Filtering 7 platforms with real-time secret alerts.
35:00 - 40:00	Q&A; & Open Source Code Release	Repo distribution & enterprise IR integration.